Copper Booms and Human Development in Zambia:

Online Appendix

Sayan Dasgupta & Pravesh Tamang

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This appendix contains all regression tables and additional figures referenced in the main text. Every table is generated programmatically from the CSV files in `OUTPUT/` by running `07_export_tables.py`; no table content is hand-typed.

Contents

- Section A: Heterogeneity and Mechanisms (Tables A1–A2)
- Section B: Robustness Checks (Tables A3–A5)
- Section C: Migration and Composition Checks (Tables A6–A8)
- Section D: Full Results by Module (Tables A9–A14)
- Section E: Additional Figures

A Heterogeneity and Mechanisms

Table A1: Urban versus Rural Heterogeneity

Outcome	Urban $\hat{\beta}$ (pp)	p	Rural $\hat{\beta}$ (pp)	p
Facility delivery	-3.90	0.151	-11.19^{***}	<0.001
Skilled birth attendance	-3.66	0.227	-13.09^{***}	<0.001
ANC 4+ visits	$+3.43$	0.191	-8.96^{***}	0.002
Postnatal care (mother)	-21.70^{***}	0.007	-22.56^{***}	<0.001
Child fever	$+5.82^{***}$	0.008	$+6.90^{***}$	<0.001
Stunted	$+16.52$	0.185	$+13.24$	0.211
Underweight	$+7.44^{***}$	0.003	$+12.70$	0.231
Male employed	$+0.83$	0.760	$+14.91^{***}$	0.004

Notes: Each cell is a separate regression on the urban or rural subsample. Province and wave fixed effects absorbed by iterative demeaning. SE clustered at province level ($G = 9$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A2: Pre versus Post Privatisation Split

Outcome	Pre-priv. $\hat{\beta}$ (pp)	p	Post-priv. $\hat{\beta}$ (pp)	p
Facility delivery	-1.32	0.998	-24.92^{**}	0.023
Skilled birth attendance	-511.09	0.604	-10.37	0.203
ANC 4+ visits	-969.13	0.234	-10.40^{**}	0.032
Postnatal care (mother)	—	—	-22.60^{***}	0.003
Child fever	-78.94	0.972	$+18.08^{*}$	0.056
Stunted	—	—	$+11.28$	0.130
Male employed	—	—	$+14.35$	0.430

Notes: Pre-privatisation sample: waves 1992 and 1996. Post-privatisation sample: waves 2007, 2014, 2018, and 2024. Wave 2002 excluded from both samples to avoid boundary overlap. Postnatal care data unavailable before 2007. Pre-privatisation point estimates for delivery outcomes are very large due to near-singular design matrices with only two treated provinces in two waves; we rely on the sign and the post-privatisation estimates for substantive interpretation. SE clustered at province level ($G = 9$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B Robustness Checks

Table A3: Wild Cluster Bootstrap versus Analytic p-values

Outcome	$\hat{\beta}$ (pp)	SE	p (analytic)	p (bootstrap)	N
Child fever	$+7.80^{***}$	0.99	<0.001	0.010	52,942
Finished floor	-6.27^{***}	1.43	0.002	0.074	69,179
Postnatal care (mother)	-22.60^{***}	5.39	0.003	0.077	19,754
Skilled birth attendance	-14.72^{***}	4.33	0.009	0.090	38,908
Facility delivery	-13.63^{**}	4.10	0.010	0.089	38,916
Choked or burned (ever)	-2.14^{**}	0.80	0.028	0.139	28,914
Has mobile phone	-15.64^{**}	5.82	0.028	0.138	27,634
ANC 4+ visits	-6.61^{**}	2.75	0.043	0.142	38,433
Male employed	$+9.22^{*}$	4.49	0.074	0.208	49,946
Owns house	$+3.32$	6.53	0.625	0.615	17,932
Justifies DV (any reason)	-4.51^{*}	2.26	0.081	0.161	58,044

Notes: Webb (2014) six-point weights, $B = 999$ replications, $G = 9$ clusters, two treated. H_0 imposed via restricted residuals. Bootstrap p-values are larger than analytic p-values because only two of nine clusters are treated, producing a coarse null distribution (MacKinnon and Webb, 2019). This is a known limitation of the bootstrap with very few treated clusters, not evidence against the main results.

Table A4: High- versus Low-Price Wave Robustness

	High-price waves		Low-price waves	
Outcome	$\hat{\beta}$ (pp)	p	$\hat{\beta}$ (pp)	p
Facility delivery	-24.92^{**}	0.023	$+13.91^{*}$	0.087
Skilled birth attendance	-10.37	0.203	$+10.34^{**}$	0.046
ANC 4+ visits	-10.40^{**}	0.032	$+7.64$	0.243
Postnatal care (mother)	-22.60^{***}	0.003	—	—
Child fever	$+18.08^{*}$	0.056	-4.10	0.590
Finished floor	-14.45	0.320	$+14.53$	0.145
Choked or burned (ever)	-2.14^{**}	0.028	—	—
Has mobile phone	-15.64^{**}	0.028	—	—
Male employed	$+14.35$	0.430	$+13.92$	0.393
Owns house	$+3.32$	0.625	—	—
Justifies DV (any reason)	$+2.90$	0.816	—	—

Notes: High-price waves: 2007, 2014, 2018, 2024 (price > \$6,500/MT). Low-price waves: 1992, 1996, 2002 (price < \$2,300/MT). Dashes indicate the outcome was unavailable or the subsample was too small. Coefficients in percentage points. SE clustered at province level ($G = 9$), $df = 8$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A5: Event Study: Wave-by-Wave Coefficients (Base = 2002)

Wave	Facility delivery	Skilled birth attend.	ANC 4+ visits	Child fever
1992	$+5.90^{**}$	$+3.35$	$+0.25$	-1.78
1996	$+6.71^{*}$	$+5.76^{*}$	$+4.28^{*}$	-1.70
2002	0 (base)	0 (base)	0 (base)	0 (base)
2007	-8.03^{**}	-9.09^{***}	-7.79^{*}	$+8.85^{*}$
2014	-11.78	-19.83^{**}	-5.07	$+8.10^{***}$
2018	-15.58^{*}	-15.54^{**}	-7.80	$+6.74^{***}$
2024	-22.31^{**}	-20.98^{**}	-10.58^{**}	$+13.15^{***}$

Notes: All coefficients in percentage points. Base wave is 2002 ($\hat{\beta}_{2002} = 0$ by construction). Province and wave fixed effects; SE clustered at province level ($G = 9$), $df = 8$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C Migration and Composition Checks

Table A6: Placebo and Composition Stability Tests

Outcome	$\hat{\beta}$	95% CI	p-value	N
Child sex ratio (Mother's age at survey (years)	$+0.27$	$[-0.32, +0.87]$	0.315	73,928
Birth order	-0.09	$[-0.31, +0.13]$	0.379	53,984
Household size (persons)	-0.16	$[-1.34, +1.03]$	0.769	69,344

Notes: Outcomes that copper prices cannot causally affect. None significant at any conventional level, supporting stability of provincial population composition across the commodity cycle. 95% confidence intervals computed from CR1S clustered SE, $t_{df=8}$ critical value. SE clustered at province level ($G = 9$), $df = 8$.

Table A7: Extended Composition Stability: Education, Marital Status, and Urbanisation Share

Outcome	$\hat{\beta}$	95% CI	p-value	N
Mean age of women (years)	+0.27	[−0.29, +0.83]	0.305	73,928
Secondary+ education (pp)	+0.73	[−2.15, +3.60]	0.576	73,916
Currently married (pp)	−1.74	[−3.95, +0.47]	0.106	73,928
Urban share (pp)	−5.52	[−32.02, +20.98]	0.644	69,344
Mean household size (persons)	−0.16	[−1.34, +1.03]	0.769	69,344

Notes: Extended composition stability test including education, marital status, and urbanisation share. None significant at conventional levels. 95% confidence intervals computed from CR1S clustered SE, $t_{df=8}$ critical value. SE clustered at province level ($G = 9$), $df = 8$.

Table A8: Long-Resident Subsample Robustness — Women (`v104` ≥ 5 or `v104` $= 95$)

	Full sample				Long-resident	
Outcome	$\hat{\beta}$	95% CI	p	N	$\hat{\beta}$	95% CI
Facility delivery	-13.63^{**}	$[-23.08, -4.19]$	0.010	38,916	-16.16^{***}	$[-26.44, -5.87]$
Skilled birth attend.	-14.72^{***}	$[-24.70, -4.74]$	0.009	38,908	-17.37^{***}	$[-28.01, -6.74]$
ANC 4+ visits	-6.61^{**}	$[-12.96, -0.27]$	0.043	38,433	-7.89^{**}	$[-14.62, -1.15]$
Postnatal care	-22.60^{***}	$[-35.04, -10.17]$	0.003	19,754	-26.09^{***}	$[-39.92, -12.26]$

Notes: Long-resident = lived in current province 5 or more years (v104 ≥ 5) or always (v104 $= 95$). Vi
who did not state duration excluded. Coefficients in percentage points. 95% confidence intervals from CR
$t_{df=8}$ critical value. SE clustered at province level ($G = 9$), $df = 8$.

Table A9: Long-Resident Subsample Robustness — Men (`mv104` ≥ 5 or `mv104` $= 95$)

	Full sample				Long-resident		
Outcome	$\hat{\beta}$	95% CI	p	N	$\hat{\beta}$	95% CI	p
Secondary+ education	-0.13	$[-2.80, +2.53]$	0.911	49,974	-0.31	$[-3.40, +2.78]$	0.821
Education level (0-3)	$+0.02$	$[-0.04, +0.07]$	0.516	49,974	$+0.01$	$[-0.05, +0.06]$	0.805
Currently employed	$+9.22^{*}$	$[-1.13, +19.56]$	0.074	49,946	$+9.35^{**}$	$[+0.13, +18.58]$	0.048
Condom at last sex	-1.71^{*}	$[-3.81, +0.39]$	0.097	38,886	-0.83	$[-3.55, +1.88]$	0.500
2+ partners last year	-0.94	$[-4.04, +2.16]$	0.503	48,099	-1.40	$[-6.06, +3.26]$	0.508
Justifies DV	-3.49	$[-12.55, +5.58]$	0.401	47,570	-3.88	$[-12.88, +5.11]$	0.348

Notes: Long-resident = mv104 $\geq$ 5 years or mv104 = 95 (always). 76.7 percent of men in the pooled M meet this criterion. Employment gains strengthen in the long-resident subsample; condom use attenuat suggesting the full-sample result partly reflects in-migration of higher-risk men. 95% confidence intervals f clustered SE, $t_{df=8}$ critical value. SE clustered at province level ($G = 9$), $df = 8$. $^{**}p < 0.05$, $^{*}p < 0.10$.

D Full Results by Module

Table A10: Full Results: Wealth and Assets

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Asset index	+0.14	[−5.01, +5.30]	0.950	69,327
Has electricity	−1.70	[−8.95, +5.56]	0.604	69,286
Has television	+0.51	[−7.41, +8.44]	0.885	69,281
Finished floor	-6.27^{***}	[−9.58, −2.96]	0.002	69,179
Improved water source	−7.11	[−28.89, +14.67]	0.473	68,908
Improved sanitation	−17.78	[−41.91, +6.35]	0.128	69,172
Has mosquito net	−4.40	[−11.05, +2.25]	0.166	55,847
Clean cooking fuel	−3.59	[−15.42, +8.24]	0.504	36,541

Notes: Controls: household size, urban residence. 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A11: Full Results: Child Health

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Stunted	+11.28	[−4.15, +26.71]	0.130	29,084
Underweight	+7.96	[−5.37, +21.30]	0.206	29,191
Wasted	+2.55	[−3.00, +8.11]	0.320	29,064
BCG vaccinated	+0.65	[−1.18, +2.49]	0.437	45,961
Measles vaccinated	−0.52	[−2.04, +1.00]	0.455	45,896
DPT3 vaccinated	−1.07	[−4.56, +2.41]	0.497	46,142
Fully vaccinated	−2.22	[−5.88, +1.45]	0.201	44,870
Diarrhoea last 2 weeks	+4.65***	[+3.05, +6.25]	<0.001	52,936
Fever last 2 weeks	+7.80***	[+5.53, +10.07]	<0.001	52,942
Cough/ARI last 2 weeks	+2.90	[−2.52, +8.32]	0.253	52,899
Ever breastfed	−0.12	[−0.65, +0.41]	0.618	50,078

Notes: Controls: child age, child sex, birth order, urban residence. Anthropometric outcomes available from 2007 onwards only. Mortality outcomes are omitted (zero within-group variation after demeaning). 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A12: Full Results: Maternal and Reproductive Health

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Facility delivery	-13.63^{**}	[−23.08, −4.19]	0.010	38,916
Skilled birth attendance	-14.72^{***}	[−24.70, −4.74]	0.009	38,908
Caesarean section	+1.01	[−1.43, +3.46]	0.368	38,804
ANC 4+ visits	-6.61^{**}	[−12.96, −0.27]	0.043	38,433
ANC first trimester	−3.04	[−8.16, +2.09]	0.209	37,766
Postnatal care (mother)	-22.60^{***}	[−35.04, −10.17]	0.003	19,754
Tetanus 2+ injections	+2.45	[−2.11, +7.02]	0.251	38,834
Iron supplementation	−2.26	[−9.87, +5.34]	0.512	30,214
Modern contraception	−3.07	[−8.39, +2.24]	0.219	73,916
Unmet need for FP	+0.36	[−6.93, +7.65]	0.912	47,132
Age at first birth	+0.25	[−0.11, +0.61]	0.143	55,358
Age at first sex	+0.32	[−1.36, +2.00]	0.671	64,656

Notes: Controls: woman's age, urban residence, married/in union, education level. 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A13: Full Results: Women's Empowerment

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Secondary+ education	+3.17	[−4.03, +10.37]	0.340	73,916
Years of education	+0.20	[−0.12, +0.52]	0.184	73,889
Currently employed	+4.91	[−4.29, +14.11]	0.253	73,835
Earns cash	+0.19	[−2.27, +2.66]	0.861	21,980
Decides own healthcare	+1.56	[−15.94, +19.06]	0.842	36,658
Decides large purchases	−0.25	[−10.38, +9.89]	0.957	36,655
Decides daily food	−4.14	[−16.19, +7.90]	0.451	36,656
Owns house	+3.32	[−11.74, +18.38]	0.625	17,932
Owns land	+3.62	[−23.80, +31.03]	0.769	13,718
Has bank account	−0.88	[−8.81, +7.06]	0.805	27,634
Has mobile phone	-15.64^{**}	[−29.06, −2.22]	0.028	27,634

Notes: Controls: woman's age, urban residence, married/in union. 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A14: Full Results: Domestic Violence

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Any violence (ever)	+3.04	[−11.01, +17.09]	0.631	29,731
Physical violence (ever)	−11.21	[−47.73, +25.31]	0.499	29,735
Emotional violence (ever)	−5.72	[−14.25, +2.80]	0.160	28,876
Slapped (ever)	−5.40	[−19.90, +9.09]	0.415	21,270
Punched (ever)	−3.25	[−8.28, +1.78]	0.174	26,928
Kicked/dragged (ever)	−2.57	[−7.92, +2.77]	0.299	26,660
Choked/burned (ever)	-2.14^{**}	[−3.99, −0.30]	0.028	28,914
Justifies DV any reason	-4.51^{*}	[−9.72, +0.70]	0.081	58,044

Notes: Controls: woman's age, urban residence, secondary education. 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A15: Full Results: Men's Outcomes

Outcome	$\hat{\beta}$ (pp)	95% CI	p-value	N
Secondary+ education	-0.13	$[-2.80, +2.53]$	0.911	49,974
Education level (0-3)	$+0.02$	$[-0.04, +0.07]$	0.516	49,974
Currently employed	$+9.22^{*}$	$[-1.13, +19.56]$	0.074	49,946
Condom at last sex	-1.71^{*}	$[-3.81, +0.39]$	0.097	38,886
2+ partners last year	-0.94	$[-4.04, +2.16]$	0.503	48,099
Justifies DV	-3.49	$[-12.55, +5.58]$	0.401	47,570

Notes: Controls: man's age, urban residence. MR files begin in 1996. See Table A9 for long-resident robustness check. 95% confidence intervals from CR1S clustered SE, $t_{df=8}$ critical value. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

E Additional Figures

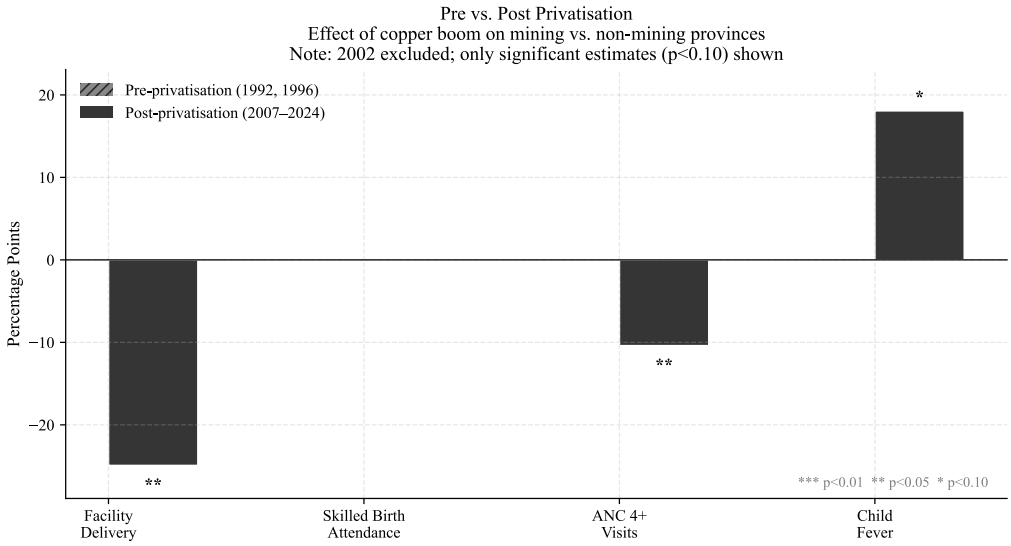

Figure A1: Pre- versus post-privatisation split: coefficient plots for facility delivery, skilled birth attendance, and child fever. Dark markers indicate post-privatisation estimates; grey markers indicate pre-privatisation estimates. The sign reversal across maternal health outcomes is the central finding of the privatisation analysis.

References

References

MacKinnon, J. G. and Webb, M. D. (2019). Wild bootstrap randomization inference for few treated clusters. *Advances in Econometrics*, 39, 65–94.

Webb, M. D. (2014). Reworking wild bootstrap-based inference for clustered errors. Queen's Economics Department Working Paper No. 1315.